COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling for Data Science

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Categorical Predictors

In this lecture:

- Learn how categorical predictors like sex, region, etc., are incorporated into a regression model
- Explain the concept of dummy coding to represent effects of levels of the categorical variables on response
- Construct dummy-coded variables for variables with two or more levels and combine with numerical predictors and interaction effects.
- Interpretation of intercept and slopes in a model with dummy-coded predictors
- Implementation with software
- Create categorical predictors from numerical
- Basis expansion regression

Example – Salaries from carData

The 2008-09 nine-month academic salary for faculty in a college in the U.S. The data were collected as part of the on-going effort of the college's administration to monitor salary differences between male and female faculty members. Stored in a data frame with 397 observations on the following 6 variables:

- `rank`: a factor with levels AssocProf AsstProf Prof
- `discipline`: a factor with levels A (“theoretical” departments) or B (“applied” departments).
- `yrs.since.phd`: years since PhD.
- `yrs.service`: years of service.
- `sex`: a factor with levels Female Male
- `salary`: nine-month salary, in dollars.
- Source: Fox J. and Weisberg, S. (2019) *An R Companion to Applied Regression*, Third Edition, Sage.

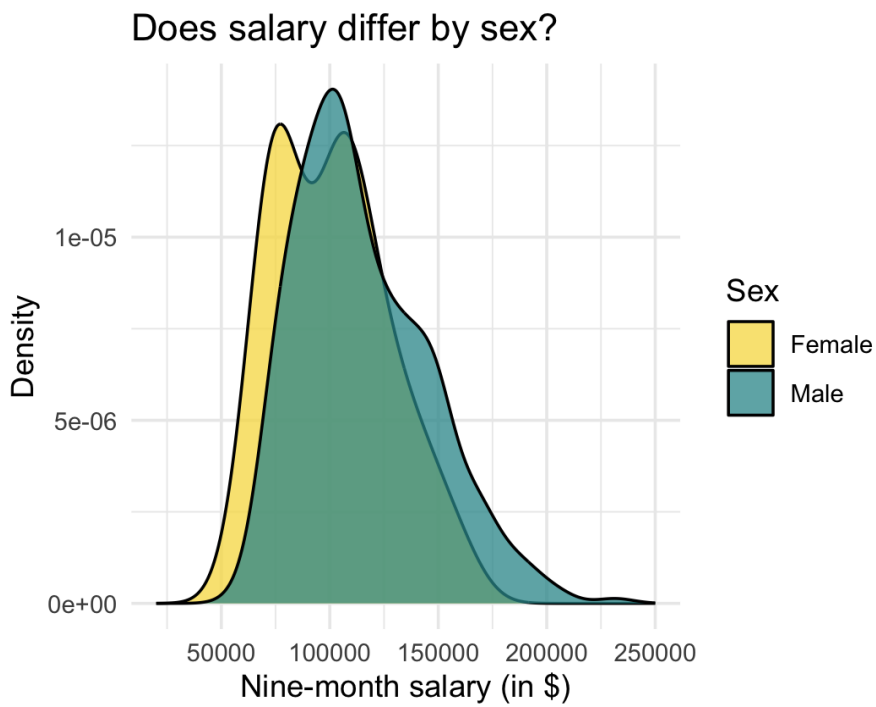
Example – Salaries from carData

Suppose that we wish to investigate differences in salaries between males and females. That is,

$$Y = \text{salary}, X = \text{sex}$$

Summary:

```
variable  sex    mean     sd    p0     p25    p50     p75   p100
salary    Female 101002. 25952. 62884 77250 103750 117002. 161101
salary    Male   115090. 30437. 57800 92000 108043 134864. 231545
```



Example – Salaries from carData

How do we fit the regression model with *Y* = salary, *X* = sex

If a categorical predictor (also known as a factor) only has two levels, or possible values, then incorporating it into a regression model is very simple. We simply create an *indicator or dummy* variable that takes on two possible dummy numerical values.

Define the following:

$$x_i = \begin{cases} 1, & \text{if } i^{\text{th}} \text{ person is female} \\ 0\,, & \text{if } i^{\text{th}} \text{ person is male} \end{cases}$$

The model is SLR:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

This is equivalent to comparing two groups:

$$y_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is female} \\ \beta_0 + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is male} \end{cases}$$

Example – Salaries from carData

The model is SLR:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

This is equivalent to:

$$y_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is female} \\ \beta_0 + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is male} \end{cases}$$

Result:

```
 term        estimate std.error conf.low conf.high
(Intercept)  115090.     1587.  111970.   118211.
female       -14088.     5065.  -24045.    -4131.
```

Interpretation:

β_0 is the average salary among males, it is estimated to be b_0 = \$115,090

$\beta_0 + \beta_1$ is the average salaryamong females, and it is estimated to be \$115,090 - \$14,088 = \$101,002

β_1 is the average gap, and is estimated to be b_1 = -\$14,088

Notice that the 95% CI for the dummy variable is all negative. This indicates that there is a statistical evidence of a difference in average salaries between the sexes.

Example – Salaries from carData

What if we use a different coding:

$$x_i = \begin{cases} 1, & \text{if i}^{\text{th}} \text{ person is male} \\ -1, & \text{if i}^{\text{th}} \text{ person is female} \end{cases}$$

This is equivalent to:

$$y_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is male} \\ \beta_0 - \beta_1 + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is female} \end{cases}$$

Result:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   108046       2532  42.667  < 2e-16 ***
sex2            7044       2532   2.782  0.00567 **
```

Interpretation:

$\beta_0 - \beta_1$ is the average salary among females, it is estimated to be \$108056 – \$7044 = \$101012

$\beta_0 + \beta_1$ is the average salary among males, and it is estimated to be \$108056 + \$7044 = \$115100

β_1 is half the gap and β_0 is the midpoint between the two average salaries.

Note: Results are the same, interpretation changes.

Example – Salaries from carData

Suppose that we wish to investigate differences in salaries between different ranks. That is,

Y = Salary, X = Rank

When a qualitative predictor has more than two levels, a single dummy variable can not represent all possible values. In this situation, we can create additional dummy variables.

$$X_1 = \begin{cases} 1 & \text{if Assistant Prof.} \\ 0 & \text{otherwise} \end{cases}$$

$$X_2 = \begin{cases} 1 & \text{if Associate Prof.} \\ 0 & \text{otherwise} \end{cases}$$

$$X_3 = \begin{cases} 1 & \text{if Full Prof.} \\ 0 & \text{otherwise} \end{cases}$$

If we include all three of the dummy variables X_1, X_2, and X_3, the problem is that the design matrix $\boldsymbol{X}$ will not be of full rank, and $\boldsymbol{X'X}$ will not be invertible. For example, 2 of each:

$$\boldsymbol{X} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Example – Salaries from carData

Define the following:

$$X_1 = \begin{cases} 1 & \text{if Assistant Prof.} \\ 0 & \text{otherwise} \end{cases}$$

$$X_2 = \begin{cases} 1 & \text{if Associate Prof.} \\ 0 & \text{otherwise} \end{cases}$$

$$X_3 = \begin{cases} 1 & \text{if Full Prof.} \\ 0 & \text{otherwise} \end{cases}$$

Solution:
Choose a "reference" category, say Category 1. Then use just the Category 2 dummy (X_2) and the Category 3 dummy (X_3)

Note: it is arbitrary which category is the reference. For example, in R, the levels are alphabetical, and the first one automatically becomes the reference.

Example – Salaries from carData

Y = Salary, $X_2 = \begin{cases} 1, & \text{if Associate} \\ 0, & \text{otherwise} \end{cases}$, $X_3 = \begin{cases} 1, & \text{if Full Prof.} \\ 0, & \text{otherwise} \end{cases}$

X_1 is not included, as Assistant Professor is the reference category (intercept)
Then the first-order regression model will be:

$$Y_i = \beta_0 + \beta_1 X_{i2} + \beta_2 X_{i3} + \varepsilon_i$$

Then depending on what values (0 or 1) we plug in for X_2 and X_3, we have:

Associate: $X_2 = 1, X_3 = 0 \Rightarrow E(Y_i) = \beta_0 + \beta_1 \times 1 + \beta_2 \times 0 = \beta_0 + \beta_1$

Full Prof.: $X_2 = 0, X_3 = 1 \Rightarrow E(Y_i) = \beta_0 + \beta_1 \times 0 + \beta_2 \times 1 = \beta_0 + \beta_2$

Assistant Professor: $X_2 = 0, X_3 = 0 \Rightarrow E(Y_i) = \beta_0 + \beta_1 \times 0 + \beta_2 \times 0 = \beta_0$

That is, the coefficients have the following meaning:

β_1 = the difference between average salaries of Associate and Assistant Prof.

β_2 = the difference between average salaries of Full and Assistant Professors

$\beta_1 - \beta_2$ = the difference between balances of Associate and Full professors

Example – Salaries from carData

Result:

$$\hat{y} = 80776 + 13100 \times X_2 + 45996 \times X_3$$

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      80776       2887  27.976  < 2e-16 ***
rankAssocProf    13100       4131   3.171  0.00164 **
rankProf         45996       3230  14.238  < 2e-16 ***
```

- Assistant professors' average salary is $80,776
- Associate professors make $14,100 more on average, compared to assistant professors
- Full professors make $45,996 more on average, compared to assistant professors
- All differences are significant.

Including both Numerical and Categorical Predictors

Suppose that we wish to predict salary using the years of service (X_1) and sex (X_2) variables, where X_2 = 1 for females.

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

$$Y_i = \begin{cases} \beta_0 + \beta_2 + \beta_1 X_{i1} + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is a male} \\ \beta_0 + \beta_1 X_{i1} + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is a female} \end{cases}$$

Interpretation:

Equivalent to fitting parallel lines to the groups.

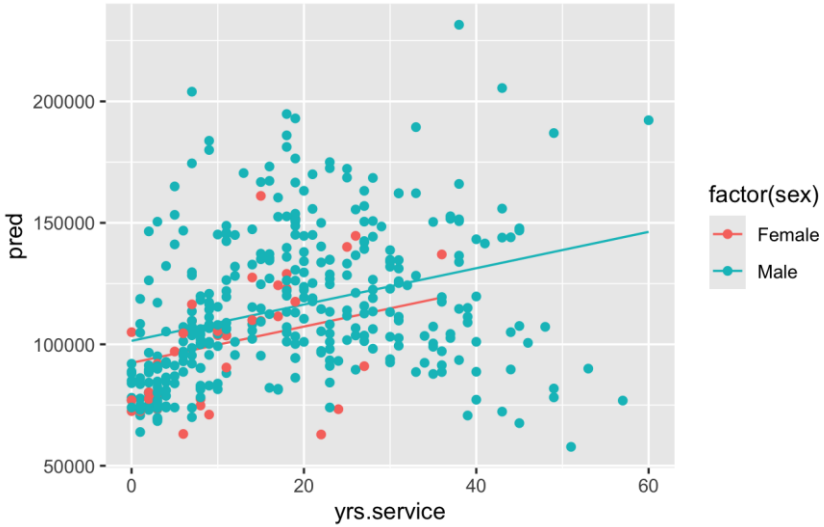
Both males and females' salaries change by β_1 dollars for each extra year of service (same slope)

However, the lines have different intercepts and β_2 is the gap between the lines.

Including both Numerical and Categorical Predictors

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  92356.9     4740.2  19.484  < 2e-16 ***
yrs.service    747.6      111.4   6.711 6.74e-11 ***
sexMale       9071.8     4861.6   1.866   0.0628 .
```

Interpretation:
Salaries increase by $747.6 on average for each extra year of service.
Females average starting salary is $92,356.90 while males average starting salary is $9,071.80 more
Q: What if we want non-parallel lines?



Interaction Model with Two Numeric Predictors

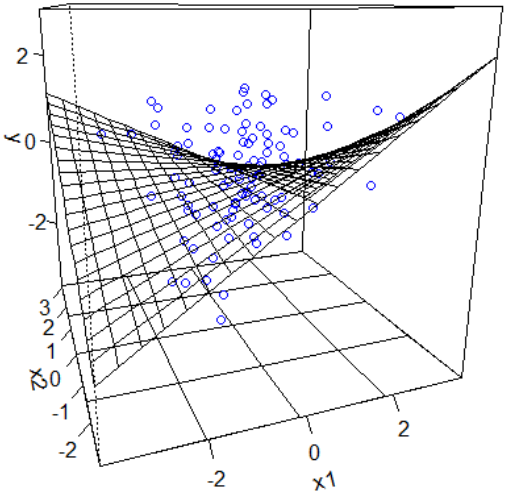
Interaction model:

$$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 X_2)$$

When $X_2 = 0$: Effect of increasing X_1 by 1: $\beta_1(1) + \beta_3(1)(0) = \beta_1$
When $X_2 = 1$: Effect of increasing X_1 by 1: $\beta_1(1) + \beta_3(1)(1) = \beta_1 + \beta_3$

That is, the effect of increasing X_1 *depends on* level of X_2, and vice versa
The geometric surface is called a "saddle".

For example,
A digital ad for sunscreen has
a high impact in the summer
when the weather is hot
but almost zero impact in the winter



Interactions Between Numerical and Categorical Predictors

- The concept of interactions applies just as well to qualitative variables or to a combination of quantitative and qualitative variables.
- We can allow the slope wrt to a quantitative Predictor to differ across levels of the Categorical Predictor

Suppose that we wish to predict salary using the years of service (X_1), sex (X_2) variables and their interaction:

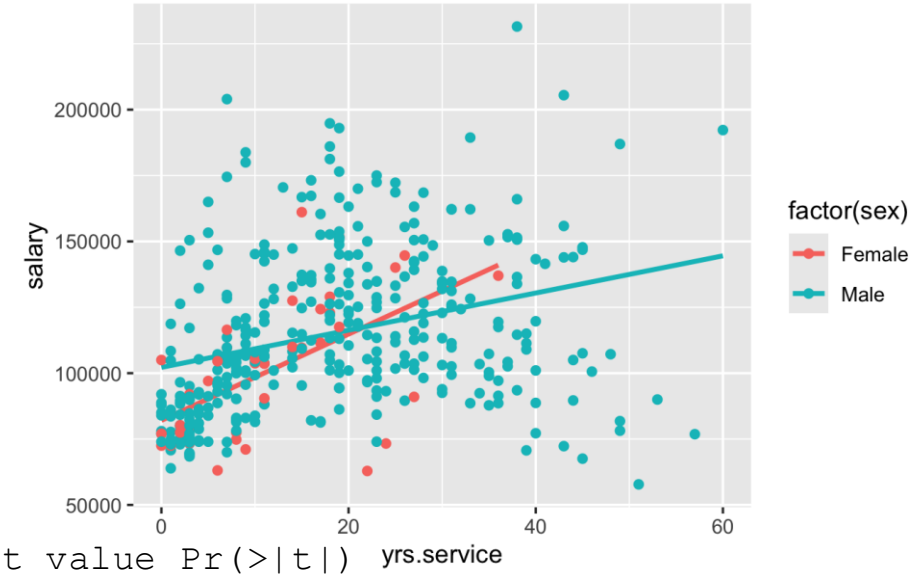
$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 (X_{i1} * X_{i2}) + \varepsilon_i$$

$$Y_i = \begin{cases} \beta_0 + \beta_2 + (\beta_1 + \beta_3) X_{i1} + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is a male} \\ \beta_0 + \beta_1 X_{i1} + \varepsilon_i & \text{if i}^{\text{th}} \text{ person is a femal} \end{cases}$$

Notice that this amounts to fitting two non-parallel lines to the data, one for females and one for males.

Interactions Between Qualitative and Quantitative Predictors

Results:

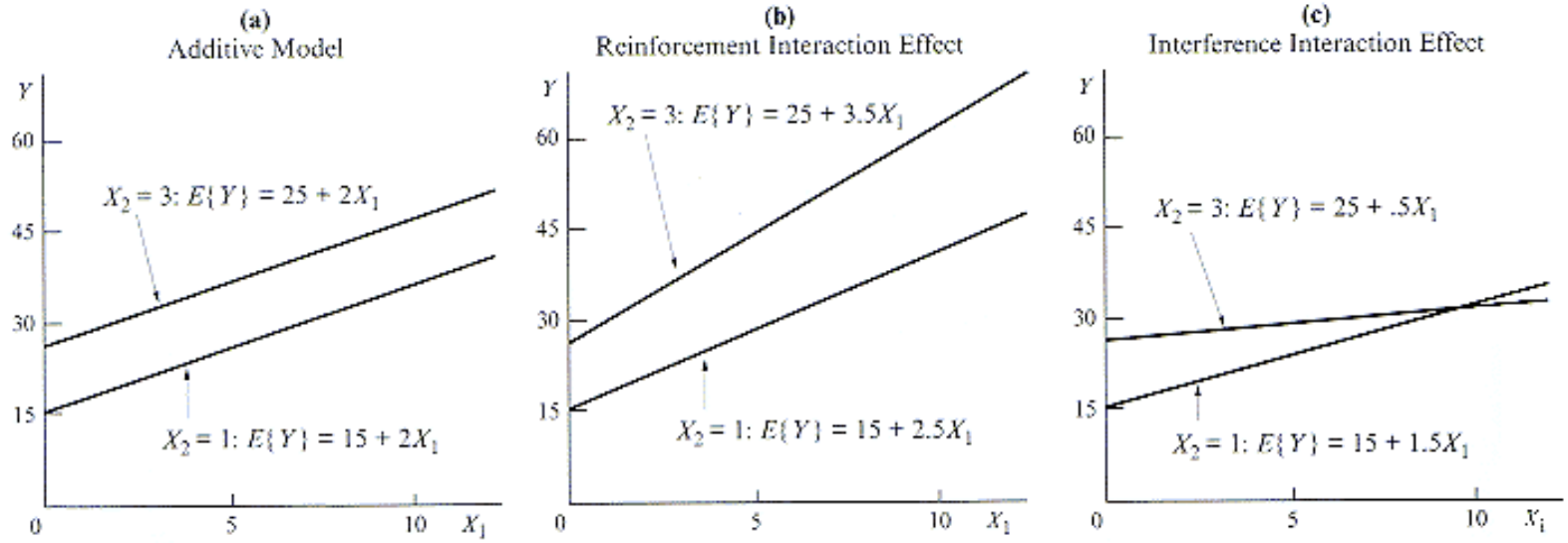



```
Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)           82068.5     7568.7  10.843  < 2e-16 ***
yrs.service            1637.3      523.0   3.130  0.00188 **
sexMale               20128.6     7991.1   2.519  0.01217 *
yrs.service:sexMale    -931.7      535.2  -1.741  0.08251 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28420 on 393 degrees of freedom
Multiple R-squared:  0.1266,	Adjusted R-squared:  0.1199
F-statistic: 18.98 on 3 and 393 DF,  p-value: 1.622e-11
```

Examples of Interaction Effects

FIGURE 7.10 **Illustration of Reinforcement and Interference Interaction Effects—Sales Promotion Example.**



Source: NKNW, p. 310

Categorical from Continuous Predictors

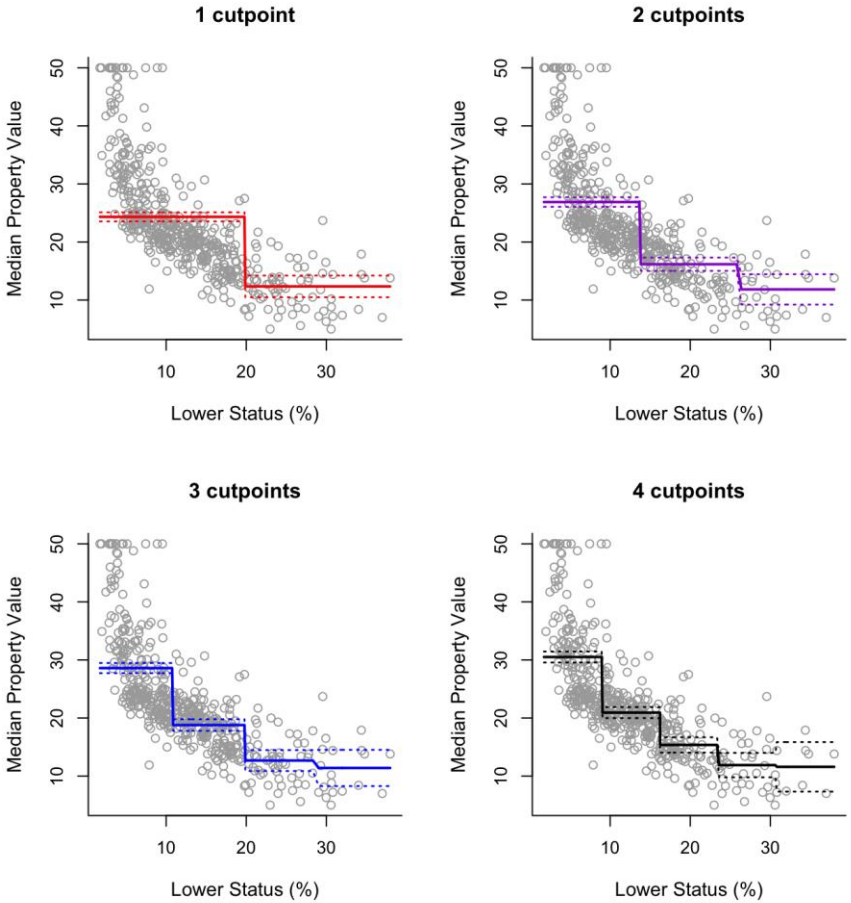
- Sometimes you might want to create a categorical predictor from a continuous.
- For example, instead of exact amount of income, you may want to split it to just three cases: low-, middle- and high-income categories.
- Why?
- Such model can handle non-linear relationships and provides more flexibility.
- These models are known as step-function or regression tree models.

Moving Beyond Linearity

- The linearity assumption is good in many statistics problems.
- However, in some cases, the true relationship between the outcome and a predictor variable might not be linear and as a result the standard linear regression can have significant limitations in terms of predictive power.
- There are other methods that offer a lot of flexibility, without losing the ease and interpretability of the linear models:
  - Polynomial regression
  - Step functions
  - Regression and smoothing splines
  - Local regression
  - Generalized additive models (GAMs)

Step Functions

- Using polynomial functions of the variables as predictor in a linear model imposes a *global* structure on the non-linear function of *X*. We need more flexibility for better predictions!
- To avoid imposing such a global structure, we can create transformations of a variable by cutting the variable range into distinct regions.
- We use *step functions* to break the range of *X* into *bins*, where we fit a different constant in each bin.
- Note: the step function regression is aka *regression tree*.



Step Functions

- This amounts to converting a continuous variable into an *ordered categorical variable*.

- In greater detail, we create *cut-points* (or knots) $c_1, c_2, \ldots, c_K$ in the range of X and then construct $K + 1$ *dummy variables*:

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \leq X < c_2), \\
C_2(X) &= I(c_2 \leq X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \leq X < c_K), \\
C_K(X) &= I(c_K \leq X),
\end{aligned}
$$

where $I(\cdot)$ is an *indicator function* that returns a 1 if the condition is true and 0 otherwise.

Step Functions

- Note that for any value of X, $C_0(X) + C_1(X) + \ldots + C_K(X) = 1$, since X must be in exactly one of the $K + 1$ intervals.

- We then use OLS estimation to fit a linear model using these $K + 1$ new variables:

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

- For a given value of X, at most one of C_1, C_2, … , C_K can be non-zero.

- β_j represents the average increase in the response for X in $c_j \leq X \leq c_{j+1}$ relative to $X < C_1$.

Tree-Based Methods

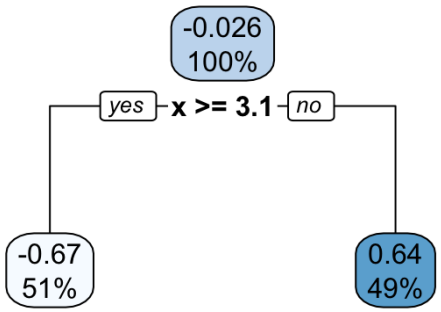
- *Tree-based* methods for regression and classification involve stratifying or segmenting the predictor space into several simple regions.

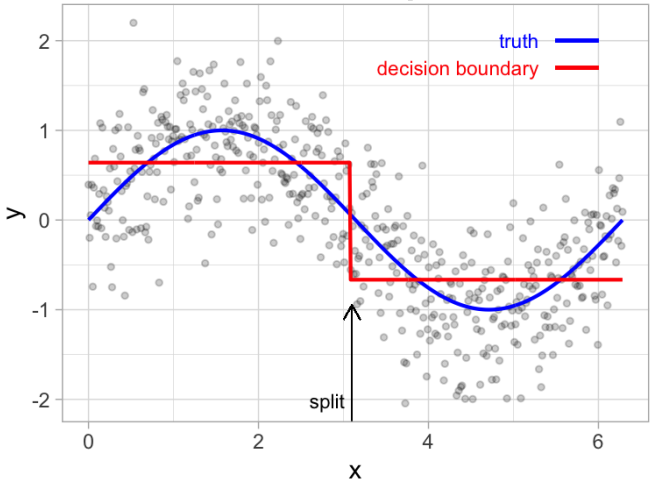
- Since the set of splitting rules used to segment the predictor space can be summarized in a tree, these types of machine learning approaches are known as *decision tree* methods.

- The basic idea of these methods is to partition the space and identify some representative centroids.

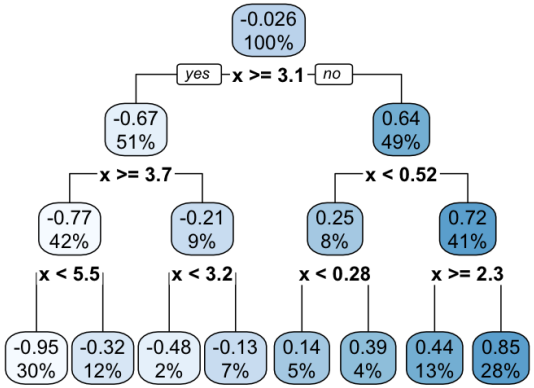
Regression Trees

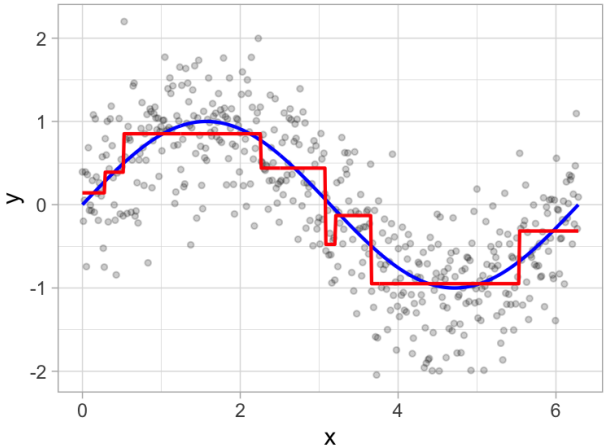
It's important to note that a single feature can be used multiple times in a tree:





If we build a deeper tree, we'll continue to split on the same feature:

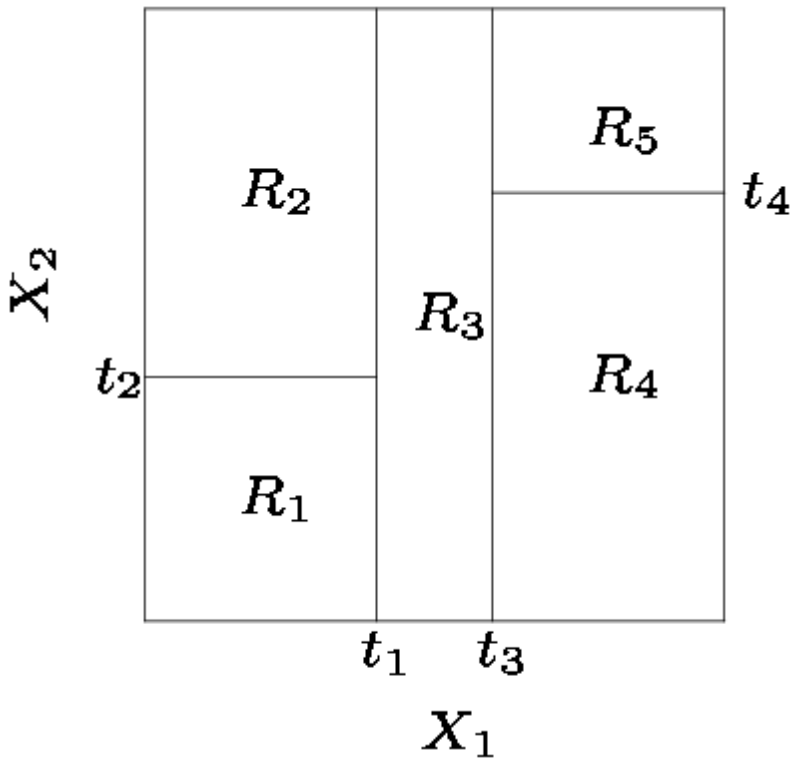




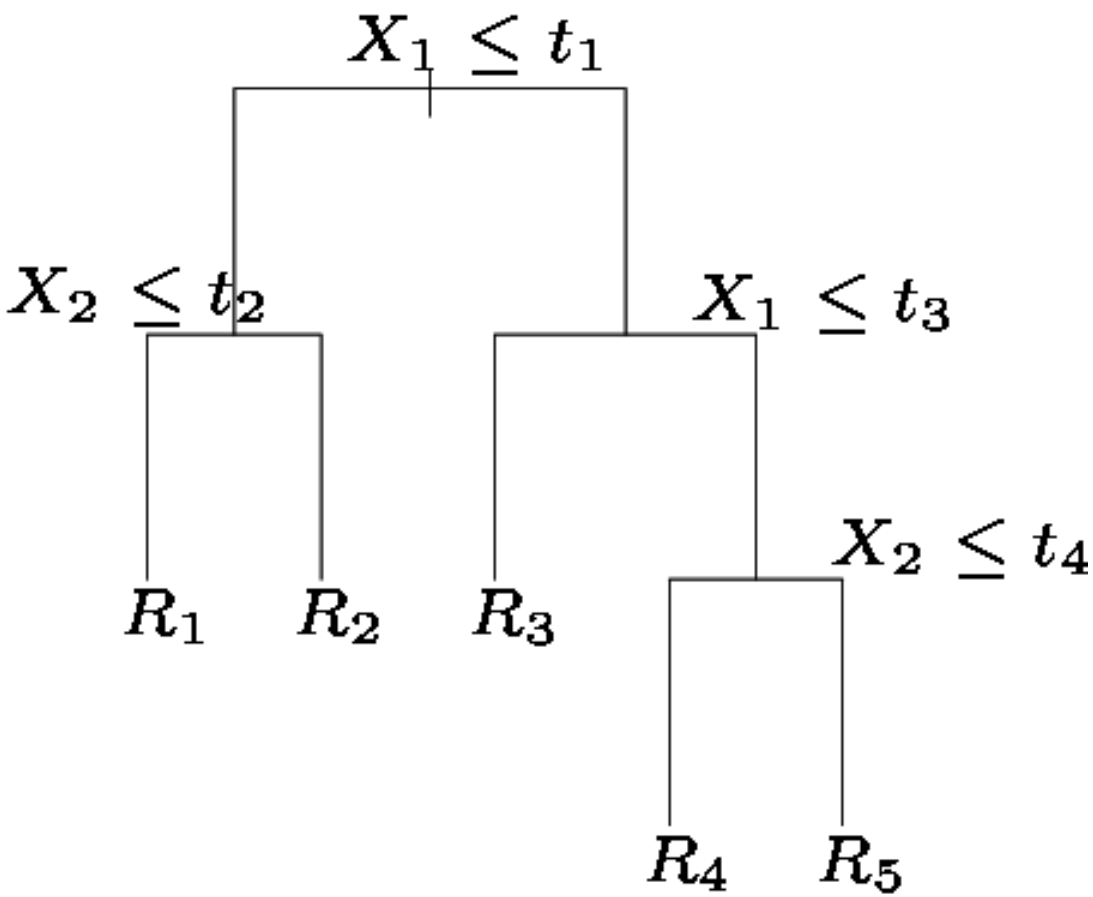
Regression Trees

- In general, we create the partitions by iteratively splitting one at a time the *X* variables into two regions.

- First split at $X_1 = t_1$
- If $X_1 < t_1$, split at $X_2 = t_2$
- If $X_1 > t_1$, split further at $X_1 = t_3$
- If $X_1 > t_3$, split at $X_2 = t_4$



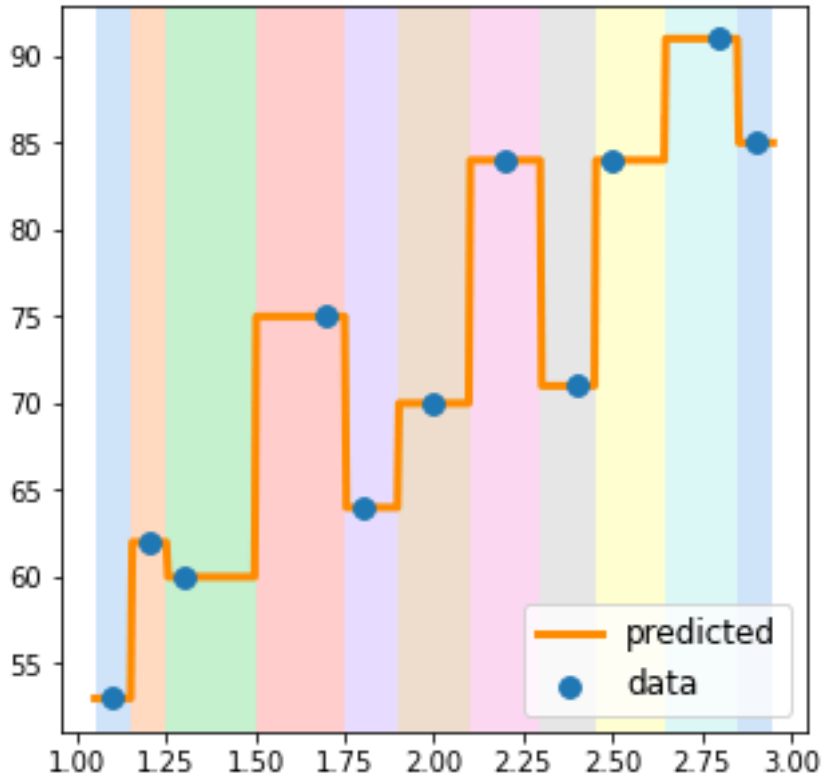
Regression Trees (cont.)



- When we create partitions this way, we can always represent them using a tree structure.
- As a result, this provides a very simple way to explain the model to a non-expert.

Regression Trees (cont.)

- Caution: Without any controls, a regression tree will grow indefinitely and overfit the data.
- One solution relies on fitting an ensemble of trees where each individual tree is regularized to be shallow. This is the random forests model.



Polynomial Regression

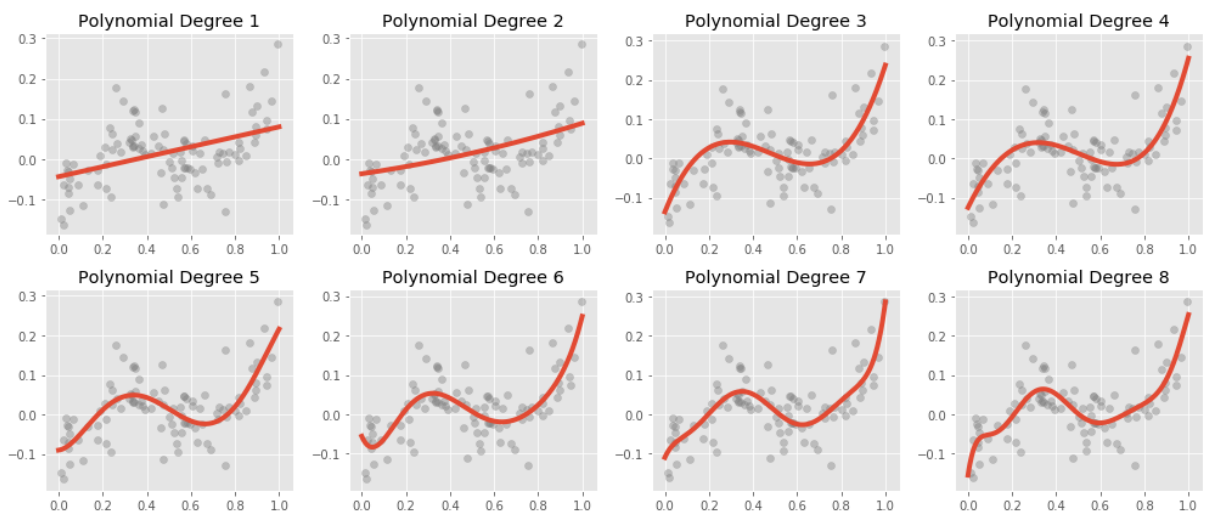
Alternative way to achieve nonlinear models:

Polynomial regression extends the linear model by adding extra predictors, obtained by raising each of the original predictors to a power. That is, replace the standard linear model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n$$

with a *polynomial* function:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d + \varepsilon_i$$



Basis Expansion

- Polynomial and piecewise constant regression models are special cases of a *basis function* approach. Each function *f*(*x*) within some class (like continuous on a finite interval), can be represented as

$$f(x) = \sum_{i=1}^{\infty} c_i b_i(x)$$

where $\{b_i(x)\}_{i=1}^{\infty}$ is an infinite collection of basis functions.

- The idea is to have at hand a family of such functions and approximate the true *f*(*x*) with only the first *K* of them.
- Instead of fitting a linear model in *X*, we fit the following model:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \ \ldots + \beta_K b_K(x_i) + \varepsilon_i$$

- Note that the basis functions $b_1(.)$, … , $b_K(.)$ are fixed and known, that is, we choose them ahead of time.

Basis Functions

- For polynomial regression, the basis functions are $b_j(x_i) = x_i^j$
- For piecewise constant functions, the basis functions are

$$b_j(x_i) = I(c_j \leq x_i \leq c_{j+1})$$

- Other examples are splines and Gaussian basis functions.
- This approach relies on math results that show most functions can be approximated with carefully selected basis functions.
- Note that we can use OLS to estimate the unknown regression coefficients.
- Thus, all the inference tools for linear models (standard errors, *F*-statistics, etc.) are available in this setting.

Piecewise Polynomials

- Instead of fitting a high-degree polynomial over the entire range of *X*, *piecewise polynomial regression* involves fitting separate low-degree polynomials over different regions of *X*.

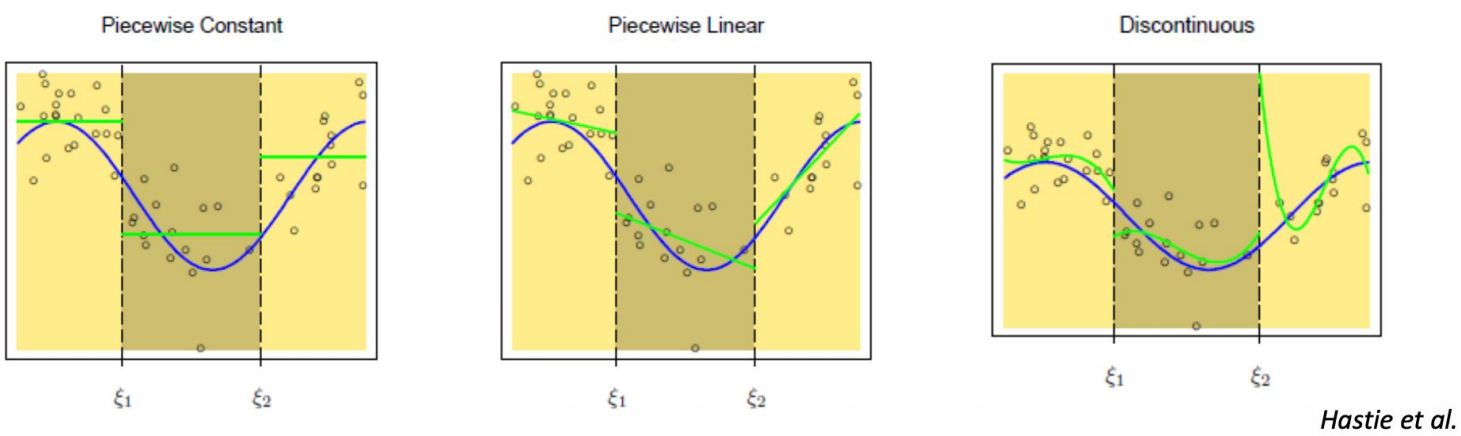
- Here the beta coefficients differ in different parts of the range of *X*; the points where the coefficients change are called *knots*.

- <u>Example</u>: A piecewise cubic polynomial with a single knot at a point *c* takes the following form:

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c; \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

Piecewise Polynomials

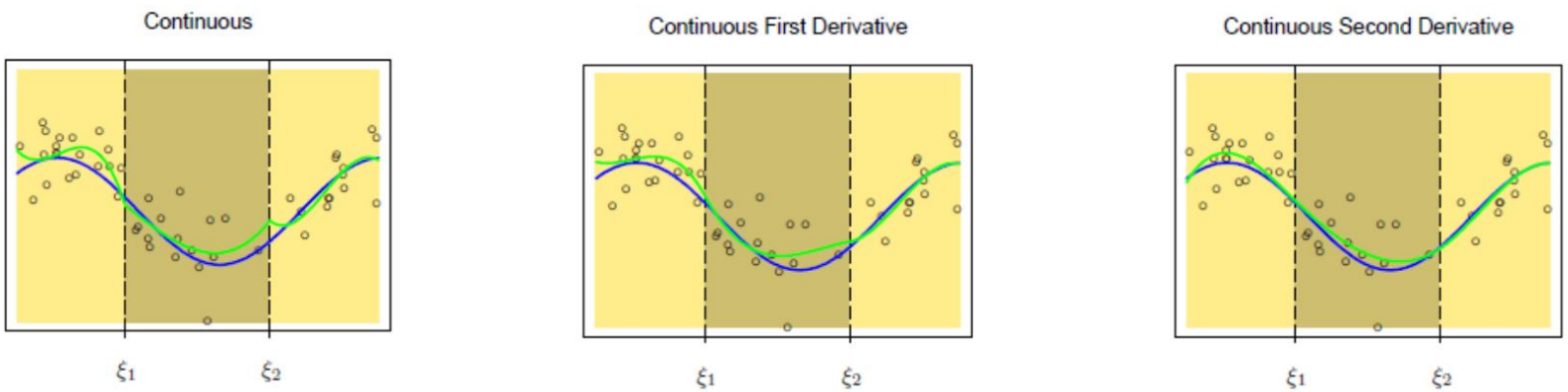
- We can continue adding higher orders of basis functions, all defined using the same knots, to obtain a piecewise polynomial fit
- For K knots and degree d polynomials, we have $(K + 1)(d + 1)$ parameters
- Issue: Fits can look strange due to discontinuities at the knots!



Hastie et al.

Smoothness Constraints

- We can impose *constraints* requiring continuity and smoothness
- Continuity requires equality of each bin's fit at each knot—K constraints
- Smoothness requires equality of *derivatives* at each knot



Regression Splines

- Splines are basis functions that satisfy a smoothness constraint.
- Degree d spline with k knots is defined as

$$f(x) = \sum_{j=0}^{d} \beta_j x^j + \sum_{k=1}^{K} \beta_{d+k} (x - \xi_k)_+^d$$

Where

$$(x - \xi)_+^d = \begin{cases} (x - \xi)^d, if\ x > \xi \\ 0, o/w \end{cases}$$

- The function f(x) is (d-1) times differentiable everywhere.